

Tide retrospective: the stabilizer and tail package (smooth-germ programme, stage C2)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-10

1 Setting

The smooth-germ programme (C2–C5) is the last gap between the formalised polynomial recovery core and germbij Theorem 3.1: the note’s own “Justification” paragraph, made formal. Its scoping consult (archived in the tide log, with the note attached as context) settled three design questions before any Lean: the fixed moving split $|x| \leq t^{-2/5}$ suffices at every finite order *provided* the Taylor power is integrated rather than sup-ed over the inner region; Taylor degree $D = R + 2$ suffices; and the stabilizer dx^M needs $M > R + 2$ — not merely $M > R$, since its scaled contribution first appears at rung $M - 2$ — with an elementary coefficient-wise formula for d .

2 The thing formalised

Three theorems in `Laplace/OneD/TaylorPackage.lean`, sorry-free with zero warnings. The centerpiece is the stabilizer envelope¹: for $a > 0$, higher coefficients at degrees $3, \dots, R' + 2$, and even M above every degree present, an explicit $d = (\sum_i |c_i| \rho^{3+i}) / \rho^M$ with $\rho = \min(1, a/(2(B + 1)))$ gives

$$\frac{a}{2} x^2 \leq a x^2 + \sum_i c_i x^{3+i} + d x^M \quad \text{for all } x,$$

by a case split at $|x| = \rho$: inside, the cubic-and-higher tail is absorbed by half the quadratic; outside, it is absorbed by the stabilizer. No suprema, no compactness. Supporting it: the exact scaling $\int |x|^k e^{-tax^2} = q^{k+1} \int |y|^k e^{-ay^2}$ under $tq^2 = 1^2$, and the tail bound³

$$\int_{|x| \geq r} |x|^s e^{-tK} \leq e^{-\frac{t}{2}\rho_K r^2} \int_{\mathbb{R}} |x|^s e^{-\frac{t}{2}\rho_K x^2}$$

for any envelope-dominated potential — the superpolynomial prefactor of the moving split, since $tr^2 = q^{-2/5}$ at $r = q^{4/5}$.

One deliberate re-scope: the partition *lower* bound requires an upper envelope near the minimum, an admissibility hypothesis that belongs with C3’s setup for the smooth loss; it moves there, and the moment upper bounds it would have carried are subsumed by scaling plus domination.

¹`exists_stabilizer_envelope`.

²`abs_moment_scaling`.

³`tail_integral.le`.

3 Honesty note

Two process events worth the record. First, this tide’s numerical check could not certify the tail rate directly: the superpolynomial regime $e^{-cq^{-2/5}} \ll q^N$ sets in only below $q \sim 10^{-6}$, so a raw check at accessible q shows *growing* values (the pre-asymptotic regime, not a counterexample); the check was replaced by its inequality-level driver, verified pointwise. Second, the first draft of the check section was written in the same shell command as the check itself and carried invented values — the session’s documented failure mode recurring — caught and corrected against the actual output before any commit, with the rule sharpened: the check runs in one command, the log text is written in a later one, quoting.

4 Mathlib vs new

Mathlib lookup: `setIntegral_mono_on`, `setIntegral_le_integral`, the linear substitution, and the even-power absolute-value lemmas. Seabed reuse: the absolute-moment integrability family. New: the stabilizer construction and the two-region assembly.

5 What the next tide inherits

Stage C3, the consult’s designated hardest step: the normalized local Taylor comparison $\langle x^s \rangle^{K_1} - \langle x^s \rangle^{K_2} = o(q^{s+D-2})$ for potentials with $K_1 - K_2 = o(|x|^D)$ at 0, from the secant bound, this tide’s tail estimate on the moving split, the inner integral with the Taylor power kept inside, and the quotient rule. It should fix the admissibility package (including the re-scoped partition lower bound) as its first act.